

Kamal Pangen, Ph.D.

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Professional Profile

I am a data scientist with 7+ years of industry experience in predictive analytics, machine learning, and GenAI. Trained as a theoretical physicist, I enjoy solving complex problems with innovative ideas. I am passionate about developing and delivering AI/ML solutions using cutting-edge technologies to enhance, augment, and automate company-wide decision-making processes.

Experience

Applied AI/ML Lead @JPMorgan Chase

May 2025-Present

Leading applied AI/ML initiatives for legal tech and eDiscovery, focusing on reducing external review costs and improving regulatory responsiveness.

- Built and enhanced eDiscovery AIML components to identify documents that are not responsive to regulatory requests, reducing the number of documents sent to outside vendors for review and saving millions in vendor spend.

Senior Data Scientist @CVS Health

Oct 2021-Present

I focus on the development and execution of ML/AI solutions to tackle fraud, waste, and abuse as well as automate the prior authorization review process.

- Developed an LLM Agent to assess members' eligibility for prior authorization approval based on electronic documents from healthcare providers. Preliminary results indicate that the agent successfully approved cases with 95% precision, streamlining the prior authorization authentication process.
- Developed a GenAI-based application utilizing the LangChain framework to expedite the creation of question sets from policy documents, reducing the turnaround time from several days to less than an hour.
- Developed anomaly detection algorithm to identify pricing irregularities in generic drugs that certain pharmacies exploited. Collaborated closely with the pricing team to swiftly rectify the issue, earning recognition with the "Premier Award" for cross-functional collaboration and rapid problem-solving.
- Developed a semi-supervised machine learning model to identify fraudulent pharmacies, employing t-SNE for pharmacy clustering and XGBoost with Shapley values to establish cluster personas. The findings from this work were forwarded to the special investigative unit for necessary action.
- Pioneered the methodology and implemented code in Teradata SQL to build the Fraud, Waste, and Abuse (FWA) data layer. This layer facilitated identifying and monitoring pharmacies and prescribers suspected of fraudulent activity.

Data Scientist - Machine Learning @Mercy Hospital, St. Louis

Feb 2018 - Sept 2021

I was responsible for the development and execution of machine learning models to automate/augment clinical workflow to drive operational efficiency and create profitability.

- Built and deployed a machine learning model to predict DRG codes of inpatient billing based on clinical notes and discrete patient data. The production model generate more than \$10 million in annual revenue by identifying and correcting incorrectly coded cases early in the billing process. This project was selected as a finalist (in the cost and efficiency category) at the company-wide innovation conference.
- Automated the cancer registry case-finding process by using NLP techniques to imaging/pathology notes and physician notes. This resulted in savings by reducing the labor cost associated with finding cancer registry cases.
- Used predictive analytics to estimate the financial impact of elective surgery cancellations due to COVID-19. The findings were presented to c-suite leadership to aid with their decision-making process.
- Developed an NLP model based on deep learning to predict the diagnosis code for inpatient admission from physician notes.

Skills

Computer Languages: SQL, Python, Mathematica, MATLAB, C++, Java, $\text{\LaTeX}$

Data analysis and visualization: BigQuery, Pandas, Matplotlib, Seaborn, QlikView, Qlik Sense, Tableau

ML / AI: Scikit-Learn, XGBoost, LightGBM, Keras, TensorFlow, PyTorch, Vertex AI, OpenAI, RAG, LLM agents, LangChain

Education

Washington University in St. Louis, St. Louis, MO

Doctor of Philosophy in Physics

AUG 2017

Masters of Science in Physics

MAY 2013

Brigham Young University, Provo, UT

Bachelors of Science in Physics and Mathematics

AUG 2011

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Machine Learning Certificates

*click on the title to view certificate

Building Basic Generative Adversarial Networks	<i>Jun 2023</i>
Causal Inference from Observational Data	<i>Jan 2022</i>
Reinforcement Learning Specialization (4 courses)	<i>Mar 2020</i>
Introduction to TensorFlow for AI, ML and DL	<i>Jun 2019</i>
Deep Learning Specialization (5 Courses)	<i>May 2019</i>
How to Win a Data Science Competition	<i>Jun 2018</i>
Machine Learning	<i>Nov 2017</i>
Awards	

Honors and Awards

• Premier Award at CVS for joining forces to better serve customers	Aug 2023
• Finalist in CVS analytic venture competition AVC2022	Nov 2022
• Finalist in Mercy wide innovation conference	Oct 2019
• Winner of 1st Mercy Hackathon [link]	Sept 2019
• Arthur L. Hughes Fellow, Washington University [link]	Summer 2012
• University Fellow, Washington University [link]	2011 & 2013
• Harvery Fletcher Scholarship, Brigham Young University	2010/2011
• Academic Scholarship, Brigham Young University	2007-2010
• Physics Department Scholarship, Brigham Young University	2007-2010

Publications

(Note: In high energy physics, name of authors are listed in alphabetical order)

- M. Alford, **K. Pangeni** and A. Windisch “Color Superconductivity and Charge Neutrality in Yukawa Theory”, Physics Review Letters 120, 082701 (2018)
- H. Nishimura, M. Ogilvie and **K. Pangeni**, “Liquid-Gas Phase Transition and CK Symmetry in Quantum Field Theories”, Physical Review D 95, 076003(2017).
- Mark Alford and **Kamal Pangeni** “Gap bridging enhancement of modified Urca process in nuclear matter.”, Physical Review C 95, 015802 (2017).
- H. Nishimura, M. Ogilvie and **K. Pangeni**, “Complex mass spectrum in lattice QCD with static quarks at strong coupling”, Physical Review D 93, 094501 (2016).
- H. Nishimura, M. Ogilvie and **K. Pangeni**, “Complex saddle points and disorder lines in QCD at finite temperature and density”, Physical Review D 91,054004 (2015).
- H. Nishimura, M. Ogilvie and **K. Pangeni**, “Complex saddle points in QCD at finite temperature and density”, Physical Review D 90,045039 (2014).